Introduction

This document gives the procedure for the estimation of measurement uncertainty (MU) for analyses in Chemistry unit.

1. Responsibilities

1. Technical Manager
2. Quality Assurance Manager
3. Laboratory Analyst

2. Identification of Contributory Factors

2.1. Identify all the contributory factors in the analysis and list them down in LF W02-CHE-01, List of Contributory Factors for Measurement Uncertainty in Chemistry.

2.2. The contributory factors can come from equipment and glassware being used, the reproducibility and calibration curve. The uncertainty can be found in the calibration certificates.

2.3. The massed up sample solution, eluate and standard stock solution, and repeated sample weight are also included in the list of contributory factors. Three repeated measurements are made for sample weight and five measurements for massed up sample solution.

2.4. The reproducibility of the analysis comes from the histamine concentration of the sample.

3. Estimation of Uncertainty of Measurement for Histamine

A. Computation of Relative Standard Uncertainty of Equipment and Mechanical Pipettes and Bottle-top Dispensers (Type B)

3.1.1Identify the measurement uncertainty of the equipment in the calibration certificate.

3.1.2. Compute for the standard uncertainty by dividing the measurement uncertainty with the coverage factor, k, which is also found in the calibration certificate. This is usually 2.

3.1.3. Compute for the relative uncertainty by dividing the standard uncertainty with the maximum weight (for analytical balances) or volume (for mechanical pipettes and bottle-top dispensers).

B. Computation of Relative Standard Uncertainty of Repeated Measurement of Sample Weight (Type A)

3.2.1. Make three measurements of sample weight.

3.2.2. Compute for the mean/average and standard deviation.

3.2.3. Compute for the standard uncertainty by dividing the standard deviation with the square root of n, which is the number of measurements.

3.2.4. The relative standard uncertainty is taken by dividing the standard uncertainty with the average weight.

C. Computation of Relative Standard Uncertainty of Repeated Measurement of Massed-up Sample Solution (Type A)

3.3.1. Fill a volumetric flask with distilled water to the mark then pour out the contents to a beaker. Measure the weight of the distilled water.

3.3.2. Make five repetitions of massed-up sample solution.

3.3.3. Compute for the mean/average and standard deviation.

3.3.4. Compute for the standard uncertainty by dividing the standard deviation with the square root of n, which is the number of measurements.

3.3.5. The relative standard uncertainty is taken by dividing the standard uncertainty with the average weight of sample solution.

D. Computation of Relative Standard Uncertainty of Glassware (Type B)

3.4.1. Identify the measurement uncertainty of the glassware which is found in the calibration certificate.

3.4.2. Compute for the standard uncertainty by dividing the measurement uncertainty with the coverage factor, k, which is also found in the calibration certificate. This is usually 2.

3.4.3. Compute for the relative uncertainty by dividing the standard uncertainty with the maximum volume.

E. Computation of Relative Standard Uncertainty of Reproducibility of Flourometric Analysis of Histamine in Sample (Type A)

3.5.1. Make three replicates of sample and compute for the concentration of histamine in sample. Refer to WI-01-CHE-02, Histamine for the analysis of histamine in fish samples.

3.5.2. Calculate the average and standard deviation of the concentration of histamine in the samples.

3.5.3. Compute for the standard uncertainty by dividing the standard deviation with the square root of n, which is the number of measurements.

3.5.4. The relative standard uncertainty is taken by dividing the standard uncertainty with the average reading of histamine concentration.

F. Computation of Relative Standard Uncertainty from calibration curve/Linearity (Type A)

3.6.1. Uncertainty from calibration curve or linearity is calculated through least-square method.

3.6.2. In order to determine the slope and y-intercept, the actual concentration of the standard solution and Relative Fluoresence Unit of the standards are placed in a cell. The LINEST function in MS Excel is used to compute for the slope. Pressing CTRL + SHIFT + ENTER will compute for the y-intercept and linear regression.

3.6.3.Use the data from the results of reproducibility of histamine for the reading (RFU) and histamine concentration in the sample. Calculate for the mean of the reading and the concentration of histamine. This will represent the x (concentration of histamine) and y-intercept (reading).

3.6.4. For the standard error of regression, type STEYX, then highlight the RFU of the standards (y values) and the actual concentration of the standards (x values).

3.6.5. Calculate for the mean of RFU of the standards and the variance of the actual concentration of standards.

3.6.6. The standard error uncertainty can be computed using the formula below:

$$Standarderroruncertainty = \frac{(SEREG)}{m} \times \sqrt{\frac{\frac{1}{Replicates\ (k)} + \frac{1}{n} + (UnknownY - MeanofY)^2}{m^{2} \times (n-1) \times Varianceofx}}$$

3.6.7. Calculate for the relative uncertainty by dividing the standard error uncertainty with the average histamine concentration of the sample (taken from the reproducibility data).

G. Computation of Overall Relative Uncertainty (U_r)

3.7.1. Calculate the overall relative uncertainty using the formula below:

$$U_r = \sqrt{{u_1}^2 + {u_2}^2 + {u_3}^2 + \cdots + {u_n}^2}$$

where u_x indicates the individual relative uncertainties that contribute to the combined uncertainty of a particular step or equipment.

H. Computation of Combined Standard Uncertainty (U_c)

3.8.1. Calculate the combined standard uncertainty using the formula below:

$$U_c = U_r \times Finalconcentrationofhistamineinsample$$

I. Computation of Expanded Uncertainty (U)

3.9.1. The expanded uncertainty can be computed by combined standard uncertainty, U_c, with the coverage factor, k (usually k=2, for a confidence level of 95%).

$$U = U_c \times k$$

This is the measurement uncertainty for the result which is expressed as:

$$(Histamineconcentration \pm U)ppm$$

4. Calculation of Measurement Uncertainty Plan

Calculation of measurement uncertainty should be done when there is new equipment, glassware or reagent brand used for the analysis, equipment has been recently calibrated and using reagent with different lot or batch number.

5. Reference

EURACHEM/CITAC Guide for Quantifying Uncertainty in Analytical Measurement, 3rd edition. EURACHEM/CITAC. 2012

6. Forms

LF W02-CHE-01, Estimation of Measurement Uncertainty for Histamine

7. Document History

Revision No.	Issue Date	Original Entry	Revised as	Recall Date